An analysis of stock returns across industries

Empirical Finance

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1 Management summary

We analyse returns on five U.S. value-weighted industry portfolios, Oil, Banks, Textiles, Toys and Retail, taken from the Kenneth French data library. We ask whether returns are predictable from the calendar month, whether the CAPM and the Fama-French five factor model price these industries and the momentum anomaly, and whether the restrictions of the classic CAPM stochastic discount factor hold.

Returns are predictable from the calendar. For every industry we reject equality of average returns across months, and the rejections survive a correction for testing five industries jointly. The differences are economically visible: September lies three to four percentage points below January for Banks, Textiles and Toys. The pattern is not stable over time, however. Across three subperiods only Textiles rejects throughout, Oil only before 1960, and Toys and Retail only after 1993, and the calendar month never explains more than two percent of monthly return variation.

The CAPM prices the five industries well, in the sense that none of the five alphas is statistically distinguishable from zero. Adding the four Fama-French factors improves the fit of the regressions but worsens the pricing: alphas for Banks, Textiles and Toys turn significantly negative, down to -0.62% per month for Textiles, because the model predicts a size and profitability premium that these industries did not earn. Neither model prices momentum, which retains an unexplained excess return of 0.71% per month under the CAPM and 0.75% under the five factor model, close to 9% per year. Sorting stocks on past returns produces a clear spread in average returns with no matching spread in factor exposures.

Estimating the classic CAPM stochastic discount factor by two-step GMM, we reject neither the over-identifying restrictions ($p = 0.31$) nor the restriction on the slope parameter ($p = 0.46$). We read this result narrowly. All five industries have market betas between 0.83 and 1.18 and similar average returns, so they give a model little to get wrong and our tests have correspondingly little power against it. The only assets in this report with genuine cross-sectional spread, the momentum-sorted portfolios, are rejected by both models. The return variation worth explaining, seasonal in the first

half of the report and momentum in the second, is captured by neither.

2 Setting the stage

In this section we will be setting the stage for the analysis and tests we will do over the U.S. stock returns data. We will first discuss the origin of the data and how they are constructed. Second, we will discuss the descriptive statistics for the industries and model factors. Lastly, we will discuss the same statistics in an overview with respect to time.

2.1 Data

We have used the data from the Kennith French Data Library. The data in this library is based on the stockmarket data from the ”Center for Research in Security Prices” (CRSP) from Fama and French [2023].

2.1.1 Industry Portfolios

We use five U.S. value-weighted portfolios, this means that bigger companies receive a higher weight in the returns of the portfolios. The specific portfolios that we have chosen are Toys, Retail, Textiles, Oil and Banks. We have chosen these portfolios such that we can look at three sectors related to household consumption and two disjoint sectors to get a better grasp of the overal economy.

2.1.2 Factor Data

In this paper we will be looking at different models each with their specific factors. .We first look at the factor used in the CAPM, which is the market excess return (Mkt-RF). The FF5 model uses the same market factor and adds four other factors: SMB, HML, RMW and CMA. The market, SMB and HML are from Fama and French [1993]

Market Excess returns: The market excess return factor (Mkt-RF) is the value weighted returns of the stock market minus the risk-free rate. It tells us something about the effectiveness of holding on to an asset from French [2026].

Small Minus Big: The small minus big factor (SMB) looks at the return difference between the portfolios of the smaller company against the bigger company. A positive SMB return implies a small company stocks returns beat big company stock returns as seen in French [2026].

High minus Low: The high minus low factor (HML) looks at the return difference between companies with high and low book-to-market equity. A positive HML factor implies that a value stocks returns outperform growth stocks returns, where value stocks are the high and growth are the low book-to-market equity as defined in French [2026].

Robust minus weak: The robust minus weak factor (RMW) is the difference between the returns of the companies with relative robust and weak profitability. If the RMW factor is positive we know that the robust firms outperform the weak companies in returns as seen in French [2026].

Conservative minus aggressive: The conservative minus aggresive factor (CMA) is about the difference between the returns of companies that invest conservative or aggressive. A positive factor tells us the the conservative companies returns outperfom the companies that aggresively invest as defined in French [2026].

Momentum: The momentum factor (MOM) is the difference between the companies that recently performed good against bad performing companies. If the MOM factor is positive the pervious good performing companies still are outperforming the bad performing companies as given by French [2026].

2.1.3 Sampling

When sampling we first remove the missing values from the data, the values we removed are noted by the -99 and -999 in the data library French [2026]. The removal of these data points is such that we could use the OLS regression without problems. After that we split the data in different time periods each specif per analysis.

2.2 Descriptive Statistics

In this section we will give a general overview of the industries and factors that we use in our analysis.

2.2.1 Industries

Before we can do an analysis of the returns of the industry portfolios is wise to look at the data. The complete descriptive statistics for all five industries are reported in Table A.1 in Appendix A. The appendix also reports the distribution of the returns in Figure A.1 and the correlation matrix in Table A.3.

Table 1: Standard Descriptive Statistics for Daily Industry Portfolio Returns

Statistic	Oil	Banks	Textiles	Toys	Retail
Mean (%)	0.05	0.05	0.04	0.05	0.05
Std. Dev. (%)	1.36	1.48	1.42	2.14	1.15
Minimum (%)	-19.93	-20.43	-22.84	-26.75	-17.97
Median (%)	0.05	0.05	0.05	0.00	0.06
Maximum (%)	19.12	19.06	19.43	39.74	17.81
Excess Kurtosis	15.03	19.37	18.57	24.88	12.97

As shown in Table 1, the first thing we notice is that the average daily weighted returns for the five industries are quite simulair, approximatly 0.05% on average across the them. The biggest difference in the mean is between Textile with 0.04% average returns against and the rest of the portfolios with 0.05%.

The bigger difference between is found when we look at the amount of risk around these returns. The most notable is the ”Toys” portfolio which is the most volitile with a standard deviation of 2.14% per day. If we set that against the least volitaile portfolio ”Retail” with a standard deviation of 1.15% on the returns, the differences become important.

When looking at the extremes of these returns we have a simulair picture. The Toys have the wides range of between the lowest and highest returns from -26.75% up to 39.74%. While retail has a lower diference between the extremes of the returns, -17.97% and 17.81%. This means that the average hides the volatility of the different portfolios

well.

The portfolios have a heavy tail. We can see this in the excess kurtosis because the values are high and positive for every industry, ranging from 12.93 to 24.88. When we look at a normal distribution, the excess kurtosis is zero. This means that these portfolios are far from idea. We observe more extremes in returns then we should when it would be normal distributed. The distributions in Figure A.1 confirm these results. The Shapiro-Wilk tests reported in Table A.1 reject normality for all industries. Which is what we expected because normality tests are sensitive to extrema.

The five portfolios are positively correlated this means that all industries are effected by market shocks. The correlation matrix can be found in Table A.3.

Overall the descriptives of these five industries tell us that the selected industries have simulair average returns but differ in volatility, extremes in returns and are sensative to market shocks.

2.2.2 Factor Data

The next portfolios we are going to describe are the factors that we use in the CAPM and FF5 models. It is useful to look at their characteristics because it gives us a better inside into the behaviour and can help us confirm our test results. The main descriptive statistics are reported in Table 2. The correlation matrix can be found in Table A while the time-series plots of the individual factors can be found in Figure A.2.

Table 2: Standard Descriptive Statistics for Monthly Factor Returns

Statistic	Mkt-RF	SMB	HML	RMW	CMA	Mom
Mean (%)	0.60	0.19	0.30	0.24	0.25	0.62
Std. Dev. (%)	4.46	3.02	2.97	2.25	2.07	4.18
Minimum (%)	-23.19	-15.54	-13.83	-18.95	-7.08	-34.36
Median (%)	1.02	0.03	0.22	0.25	0.09	0.72
Maximum (%)	16.10	18.46	12.86	13.05	9.01	18.05
Excess Kurtosis	1.73	3.06	2.18	10.57	1.34	9.76

As shown in Table 2, the average monthly returns of all the factors are a positive but a lot more diverse. When looking at different specific industry portfolios it is quite

clear that a positive return tells us something positive about the returns in the specific markets of those industries. The factors for CAPM and FF5 only tell us something about the relationship inside the specific market.

The market factor Mkt-RF is a positive value of 0.60%, this means that the market outperformed the risk-free rate. For the remaining factors, we see that a positive SMB return of 0.19% on average means that small firms outperformed large firms, while a positive HML return means that value stocks outperformed growth stocks. Similarly, a positive RMW of 0.24% returns on average shows that firms with robust profitability perform better then weak-profitability companies. While a positive CMA return means that conservative-investment companies outperformed aggressive-investment companies. Lastly for momentum, a positive return means that recent winners continued to outperform recent losers on average over the months.

What we have to keep in mind is that because we have relative high volatility for each factor, ranging from 2.07% to 4.46% that for example the small companies do not always outperform the bigger companies. When looking at the extremes we notice that they have lower lows then higher highs thus even tho we have a positive average the volatility likes so sway harder on the negative side.

All factor have positive excess kurtosis this tells us again that there is a heavier tail then if they were normally distributed. The tail is especially strong with *RMB* and *Mom* with a value of 10.57 and 9.76. The positive excess kurtosis is also consistent with the non-normal behaviour of the factor returns.

The factors are not perfectly correlated. As shown in Table A.4, the strongest positive correlation among the Fama-French factors is between HML and CMA which lies around the 0.68. While the correlation between SMB and HML is close to zero. This implies that these factors capture different kind of return variations.

Overall, we have seen the difference between the factor statistics against the industries. The volatility, tail behaviour, and economic interpretation are also a lot

different and this is what we will use when interpreting the factor regressions later in the analysis.

2.3 Time-Series Behaviour

The descriptive statistics give us a good summary of the distribution of our portfolios, but they do not show us how return variation change over time.

2.3.1 Industries

In this subsection we will take a look at this variation. We focus mainly on the Toys and Retail time-series, while the time-series plots for all five industries can be seen in Figure A.2.

When we look at the behaviour of our distribution over time in Figure 1 we notice that in the early years of our portfolio there is a lot more volatility. This is inline with the

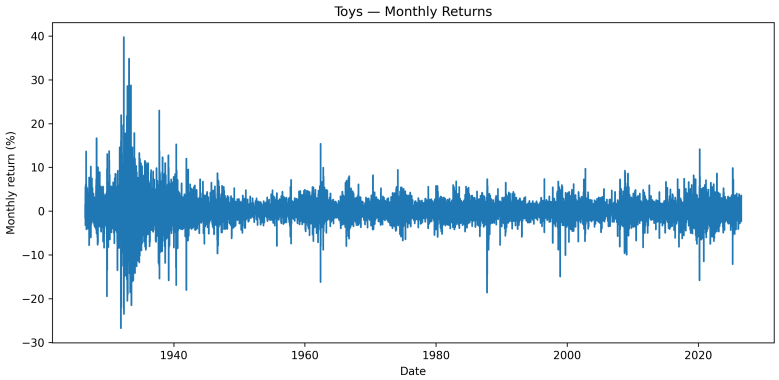


Figure 1: Toys

descriptive statistics only we notice that its volatility slows down over time. Compared to the volatility of the Retail portfolio in Figure 2 we notice that these observations do not match. The volatility does not change as much as the toys and stays somewhat more steady over time.

Overall, the time-series plots of all the industries in Figure A.2 share some characteristics. There are some calm periods and rough periods where the return fluctuations overlap. These periods have some overlap in the industries which is what we expected from the correlations, the so called sensitivity to market shock that we named. However the magnitude of the shockwave differ a lot per period and industry. The Toys portfolio is the one that shows the most reduction of volatility over time. The rest of the portfolios do not change drastically in distribution over time.

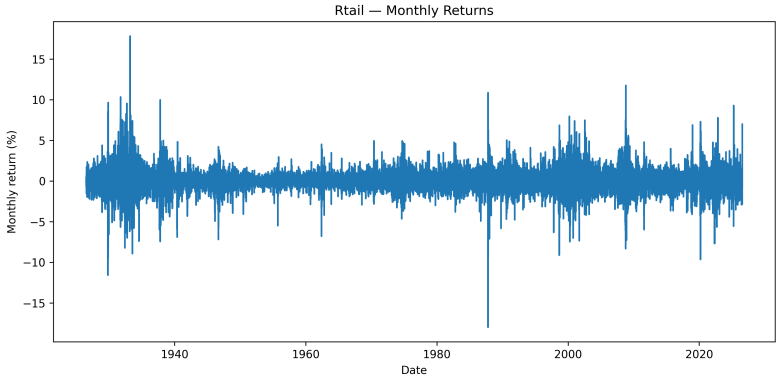


Figure 2: Retail

2.3.2 Factor Data

In this section we will discuss the factor returns over time. We will focus mainly on $Mkt - Rf$ and CMA because they are the most and least volatile of the FF5 and CAPM factors. The time-series plots of the SMB, HML, Mom and RMW factors can be found in Figure A.3.

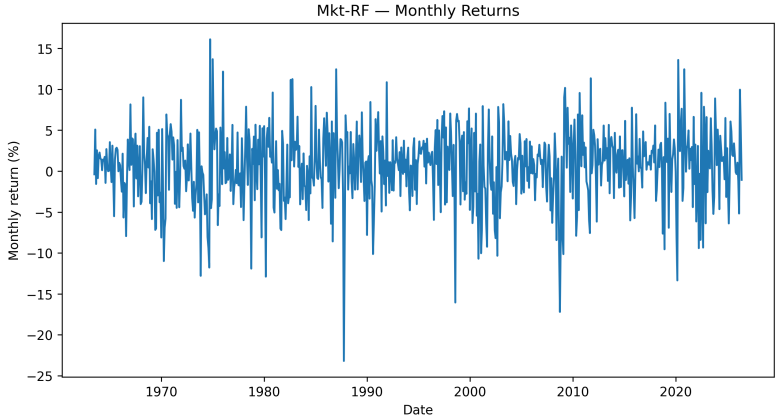


Figure 3: Monthly Market Excess Returns

Figure 3 shows us that the Mkt-RF time-series has a lot of variation over time. This means that al tho the risk-free rate outperforms the market on average that there is a lot of volatility, which causes a lot of extrema. This is also what we found in the descriptive statistics. Where we first saw a lot of calm and rough periods with a shock effect in multiple industries is that not the case anymore. The factors are relative measures of returns thus these effects cannot be seen in the same light. The desciptive statistics confirm that there is no shock effect when looking at the correlation between these factors in Table A.3.

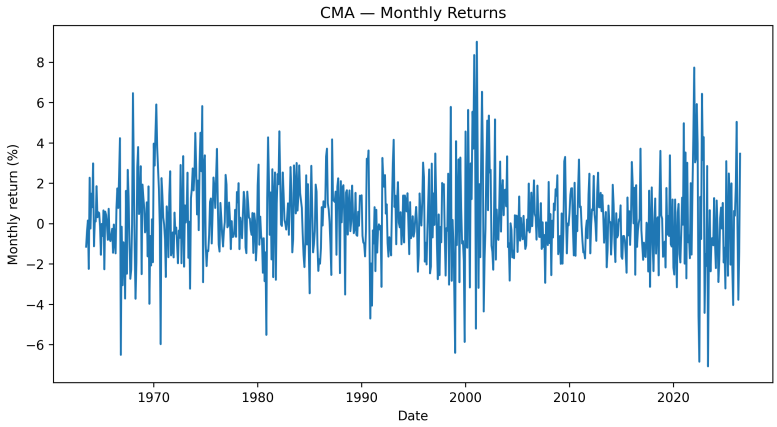


Figure 4: Monthly CMA Returns

When we compare the Mkt-RF with the CMA returns in Figure 4 we notice that there is a lot less volatility in the CMA returns. Most CMA observations remain relatively close to zero, which is consistent with its lower standard deviation of 2.07%. This tells us that conservative or aggressive strategies do not differ that much on the returns they get. In general over time a conservative strategy is better because of the positive mean but the time periods in itself may tell you if it is at the moment a good strategy.

Overall, the time-series confirm that the factor returns are changing a lot over time and that their volatility is a lot more spread out compared to the industries. It also shows us that we have to interpreted the factors a lot more with respect to their period. Because we cannot look directly at the market changes and see the effects in the factors.

3 Predictability

Efficient markets rely on traders taking arbitrage opportunities at such speed that arbitrage opportunities become infeasible, so any time variation in expected returns should be traded away. Thus, to test the efficient market hypothesis, we test whether expected returns differ for each month.

3.1 Month-of-the-year Model

The dataset consisted of monthly returns across 5 industries: Oil, Banks Textile, Toys and Retail and the equation is as follows:

$$R_{i,t} = \alpha_i + \sum_{m=2}^{12} \delta_{i,m} D_{m,t} + \varepsilon_{i,t}, \tag{1}$$

$$D_{m,t} = \begin{cases} 1, & \text{if observation } t \text{ occurs in month } m, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

$$\begin{aligned} &H_0 : \delta_{i,2} = \delta_{i,3} = \cdots = \delta_{i,12} = 0, \\ &H_1 : \text{at least one } \delta_{i,m} \neq 0. \end{aligned} \tag{3}$$

$$\mathcal{I} = \{\text{Oil}, \text{Textiles}, \text{Retail}, \text{Banks}, \text{Toys}\}, \qquad i \in \mathcal{I}. \tag{4}$$

To test for general month-of-the-year effects, we estimate Equation 1 separately for each industry.

January is omitted and therefore serves as the reference month. Thus, α_i represents the expected return of industry i in January, while its expected return in month m is $\alpha_i + \delta_{i,m}$. Consequently, each coefficient $\delta_{i,m}$ measures the difference between the expected return in month m and the expected return in January. We test the joint hypothesis in Equation 3. Rejection of the null hypothesis provides evidence that average returns differ systematically across calendar months.

To choose the standard errors, we test for heteroskedasticity and residual autocorrelation. To test for heteroskedasticity, we use the ARCH-LM test, since it is common for there to be volatility clustering in financial returns. Indeed, across all 5 industries, we

reject the null hypothesis of no ARCH effects. The test shows that the error variance varies over time, and we have ARCH type volatility clustering. The choice is thus restricted between HAC and HC standard errors
as seen in Schwert [2003], which also tests for time variation within results, uses HC. He argues that, under maintained null hyposthesis of efficient markets,

$$\mathbb{E}\left[\varepsilon_{i,t+1} \mid \mathcal{F}_t\right] = 0, \tag{5}$$

meaning that on time t, the expected error for t+1 = 0 for all t. Under this maintained assumption, Eicker-Huber-White standard errors are sufficient.
Rather than imposing serial independence, we test for residual autocorrelation using the Breusch-Godfrey test at lags 12, 24 and 36. We reject the null hypothesis of no serial correlation in the regression residuals for all industries except Oil. For the other 4 industries, we reject the null at lag 24 months. We use 24 as the bandwidth for HAC standard errors across all portfolios, to permit comparable inference. We also select different bandwidths for robustness

Table 3 provides evidence of month-of-the-year effects in all five portfolios of industries. The joint Wald test rejects the null hypothesis that the eleven February–December coefficients are jointly equal to zero for every industry. The unadjusted p-values range from 0.0001 for Textiles to 0.0172 for Oil. All five tests remain significant after applying the Holm correction across industries, with adjusted p-values below 0.05. Hence, the results indicate that average returns vary systematically throughout the calendar months and are not solely due to the increased probability of false rejection that arises from testing multiple industries. The statistical evidence is strongest for Textiles, followed by Toys, Retail, Banks, and Oil.
These results provide evidence against the strict hypothesis that expected returns are constant across all months. Since the calendar month is known in advance, seasonal patterns constitute evidence against market efficiency, and monthly differences may reflect compensation for systematic risk. It is worth noting that, for the bandwidths of 6 and 12 we do not reject the null hypothesis of no seasonality effects for oil industry. For all other industries, across all used bandwidths, we reject this hypothesis. It is important to note that we use returns, not adjusted returns, as an independent variable.

Table 3: Month-of-the-Year Regressions for Industry Returns

	Oil	Banks	Txtls	Toys	Rtail
January mean	0.624	2.546***	2.623***	2.167**	1.386**
	(0.594)	(0.899)	(0.746)	(0.882)	(0.583)
February − January	-0.182	-1.358	-1.192	0.005	-0.236
	(0.704)	(0.994)	(0.856)	(1.144)	(0.702)
March − January	0.969	-1.757	-2.796**	-0.625	0.230
	(0.948)	(1.199)	(1.122)	(1.195)	(0.816)
April − January	2.075**	-1.023	-0.506	-0.728	0.041
	(0.955)	(1.096)	(1.446)	(1.183)	(0.847)
May − January	-0.180	-1.872	-2.216**	-1.389	-0.576
	(0.755)	(1.251)	(0.972)	(1.110)	(0.747)
June − January	0.100	-1.761**	-2.357***	-2.416**	-0.123
	(0.756)	(0.838)	(0.866)	(1.088)	(0.738)
July − January	0.808	-0.751	-0.725	-1.452	0.245
	(0.810)	(1.154)	(1.235)	(1.392)	(0.917)
August − January	0.273	-0.864	-1.496	-0.308	0.114
	(0.824)	(0.962)	(1.183)	(1.058)	(0.925)
September − January	-1.115	-3.334***	-3.867***	-3.031**	-2.081**
	(0.914)	(1.265)	(1.102)	(1.220)	(0.925)
October − January	0.501	-1.892	-3.455***	-2.030	-1.057
	(0.863)	(1.216)	(1.070)	(1.460)	(0.935)
November − January	0.825	-0.726	-1.166	0.401	0.926
	(0.882)	(1.202)	(1.051)	(1.258)	(0.917)
December − January	0.757	-1.256	-0.888	-3.696***	-1.335*
	(0.735)	(1.077)	(1.065)	(1.281)	(0.802)
Joint Wald statistic	23.076	24.750	37.062	28.898	25.729
Joint p-value	0.0172	0.0099	0.0001	0.0024	0.0071
Holm p-value	0.0214	0.0214	0.0006	0.0094	0.0214
Observations	1188	1188	1188	1188	1188
R^2	0.0139	0.0131	0.0214	0.0183	0.0173

Notes: The dependent variable is the monthly value-weighted industry return, expressed in percentage points. January is the omitted reference month, so the intercept is the average January return. The remaining coefficients measure the difference between the corresponding month's average return and the January average. Newey-West HAC standard errors with a 24-month bandwidth are reported in parentheses. The joint Wald test examines whether all eleven February–December coefficients are jointly equal to zero. Holm-adjusted p-values account for the five industry-level tests. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Thus, it could be that this variance in expected returns could be due, for example, to variation in the interest rate, rather than extra compensation for taking the same risk

3.2 Sub-period Analysis

To examine the temporal robustness of the results, the full sample is divided into three equally sized periods: 1927-1959, 1960-1992, and 1993–2025. Each sub-period contains 396 monthly observations, providing 33 observations for every calendar month. The month-of-the-year regressions are re-estimated using Newey-West HAC standard errors, and the resulting industry-level p-values are adjusted using the Holm correction within each sub-period.

Dividing the data into chronological sub-periods allows us to examine whether calendar effects are stable over time.

This is important because the full sample covers almost a century, during which market liquidity, transaction costs, regulation, and industry portfolios may all have changed substantially. It is also possible that time invariance in expected returns have already been traded away.

Thus, sub-period analysis helps determine temporal stability and present-day relevance of the full sample results. It is important to note that, since we are now dealing with fewer observations per analysis, we lose statistical power, which of course makes it harder to reject the null hypothesis. The p-values across the industries and time periods are in table 4 while the table with the full coefficients can be found in the appendix:

Table 4: Holm-adjusted p-values of Month-of-the-Year Effects Across Subperiods

Industry	1927-1959	1960-1992	1993-2025
Oil	**0.0481**	0.1586	0.1264
Banks	0.5541	0.1586	0.0679
Textiles	**0.0006**	**0.0004**	**0.0045**
Toys	0.5541	0.1586	**0.0263**
Retail	0.5541	0.1586	**0.0003**

Notes: The table reports Holm-adjusted p-values from the joint Wald test of the eleven February-December coefficients. The Holm correction is applied across the five industry-level tests within each sub-period. Each sub-period contains 396 monthly observations. Newey-West HAC standard errors are calculated using a 24-month bandwidth. Values in bold indicate rejection of the null hypothesis of no month-of-the-year effects at the 5% significance level.

3.3 Results

Table 4 shows that the full-sample results are not equally robust across time. Textiles is the only industry with significant month-of-the-year effects in all three sub-periods, providing the strongest evidence of persistent calendar predictability. Oil displays significant seasonality only during 1927-1959, whereas Toys and Retail exhibit significant effects only during 1993–2025. Banks is significant in none of the three sub-periods at the 5% level, although its result for 1993–2025 is marginal. Thus, the seasonal patterns generally vary over time, suggesting that most of the full-sample results are not stable throughout the entire sample. However, the weaker sub-period evidence may also partly reflect reduced statistical power because each sub-period contains fewer observations.

Nevertheless, one could hypothesize why, for instance, time-variant expected returns are more likely to persist in industries such as textiles than in Oil or Banking. The textile industry may be more heterogeneous across the world, with more firms competing in less liquid markets which may increase trading costs and lower liquidity and thus limit the ability of arbitrageurs to eliminate predictable seasonal return patterns. Furthermore, it could be that oil and banks are more susceptible to geopolitical events, making monthly effects less stable and more difficult to precisely estimate. However, these explanations remain hypotheses.

4 Factor models

In this section we look at different factor models and how they explain the returns of the five industry portfolios and the momentum factor. We first discuss the assumptions we need for the time-series regressions, after which we test the CAPM and the Fama-French five-factor model. Lastly, we look at the momentum portfolios to see if the differences in expected returns are reflected in their factors.

4.1 Assumptions

This section analyzes historical regression data from July 1, 1963, to June 1, 2026. Because this is a time-series dataset, t-statistics are calculated using Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors.
Because of this there are some relaxations for the OLS assumptions. In particular Homoskedasticity and uncorrelated errors don't have to hold anymore, instead we now need Stationarity and Ergodicity assumptions.

Linearity in Parameters: The excess returns of the portfolios are linearly related to the factors:

$$\begin{aligned} R_{i,t} - R_{f,t} = \alpha_i + \beta_{\text{Mkt-RF},i}(R_{M,t} - R_{f,t}) + \beta_{\text{SMB},i}\text{SMB}_t + \beta_{\text{HML},i}\text{HML}_t \\ + \beta_{\text{RMW},i}\text{RMW}_t + \beta_{\text{CMA},i}\text{CMA}_t + \varepsilon_{i,t} = \gamma'\mathbf{F}_t + \varepsilon_{i,t} \end{aligned} \tag{6}$$

Strict Exogeneity: The error term has a conditional expectation of zero given the factor returns across all time periods:

$$\mathbb{E}[\varepsilon_{i,t} \mid \mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_T] = 0 \quad \forall\, t = 1, \ldots, T$$

This guarantees that the OLS parameter estimates $\hat{\alpha}_i$ and $\hat{\boldsymbol{\beta}}_i$ are unbiased and consistent.

No Perfect Multicollinearity: The factor matrix $\mathbf{F}$ has full column rank K:

$$\text{Rank}(\mathbf{F}'\mathbf{F}) = K \quad \Longleftrightarrow \quad \det(\mathbf{F}'\mathbf{F}) \neq 0$$

This guarantees that no factor is a perfect linear combination of the other factors, allowing the parameter matrix $(\mathbf{F}'\mathbf{F})^{-1}$ to be uniquely inverted.

Ergodicity for Standard Errors: The process governing the moment conditions $\mathbf{g}_t = \mathbf{F}_t\varepsilon_{i,t}$ is ergodic, ensuring that time averages converge to population expectations:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbf{g}_t\mathbf{g}_t' \xrightarrow{p} \mathbb{E}[\mathbf{g}_t\mathbf{g}_t'] \quad \text{as } T \to \infty$$

This guarantees that sample variance estimators converge to constant parameters, allowing for valid standard errors and hypothesis testing.

Stationarity for Standard Errors: The moment process $\mathbf{g}_t = \mathbf{F}_t\varepsilon_{i,t}$ is covariance-stationary, ensuring that its mean, variance, and autocovariances are invariant over time:

$$\mathbb{E}[\mathbf{g}_t] = \mathbf{0} \quad \text{and} \quad \text{Cov}(\mathbf{g}_t, \mathbf{g}_{t-k}) = \mathbf{\Gamma}_k \quad \forall\, t$$

This guarantees that the underlying population standard errors remain constant targets rather than time varying parameters.

4.2 6 time series tests of CAPM

A time-series test of the Capital Asset Pricing Model (CAPM) was conducted for each of the five industry portfolios. The following regression was estimated individually for each industry $i \in \{1, 2, 3, 4, 5\}$, corresponding to the Oil, Banks, Textile, Toys, and Retail portfolios, respectively:

$$R_{i,t} - R_{f\,t} = \alpha_i + \beta_{\text{Mkt-RF},i}(R_{M,t} - R_{f,t}) + \varepsilon_{i,t}$$

Additionally, a CAPM regression was performed on the Momentum factor (Mom) obtained from Kenneth French's data library to test whether the model accounts for the momentum effect. The Momentum factor is constructed monthly from the intersection of two size-sorted portfolios (split at the NYSE median market capitalization) and three prior-return portfolios (split at the 30th and 70th NYSE percentiles based on returns from month $t-12$ to $t-2$), giving us:

$$\text{Mom}_t = \frac{1}{2}(\text{Small High} + \text{Big High}) - \frac{1}{2}(\text{Small Low} + \text{Big Low})$$

To evaluate whether the market factor captures the momentum anomaly, we estimate the following CAPM regression:

$$\text{Mom}_t = \alpha_{\text{Mom}} + \beta_{\text{Mkt-RF,Mom}}(R_{M,t} - R_{f,t}) + \varepsilon_{\text{Mom},t}$$

Table 5: CAPM Regression Results

Industry Portfolios	**Mean Monthly Excess Return (%)**	α_i	**t-stat** α_i	$\beta_{\textbf{Mkt-RF}\ i}$	**t-stat** $\beta_{\textbf{Mkt-RF}\ i}$	**Adj** R^2
Oil	0.6662	0.1650	0.9336	0.8342	21.2467	0.3737
Banks	0.6090	-0.0379	-0.2800	1.0767	35.7611	0.6286
Txtls	0.5705	-0.1269	-0.6182	1.1609	25.4420	0.4612
Toys	0.4218	-0.2888	-1.5434	1.1828	28.4416	0.5169
Rtail	0.7360	0.1346	1.1998	1.0010	40.1646	0.6811
Momentum Factor	**Mean Monthly Excess Return (%)**	α_{Mom}	**t-stat** α_{Mom}	$\beta_{\textbf{Mkt-RF}\ Mom}$	**t-stat** $\beta_{\textbf{Mkt-RF}\ Mom}$	**Adj** R^2
Momentum	0.6215	0.7141	4.7170	-0.1541	-4.5796	0.0258

The results of these regressions are presented in Table 5. As shown, α_i is not statistically significant for any of the industry portfolios, which aligns with the predictions of the CAPM. Furthermore, the market beta ($\beta_{\text{Mkt-RF},i}$) is statistically significant across all industry portfolios, indicating that these sectors are exposed to systematic risk and are sensitive to market excess returns.

Among these portfolios, Toys exhibits the highest market beta suggesting greater sensitivity to market downturns, possibly because consumers cut back on non-essential spending during market downturns. Conversely, Oil displays the lowest market sensitivity, which might be explained by fuel and energy being basic necessities. Finally, Table 5 shows that the intercept for the Momentum factor is positive and statistically significant. This confirms that the standard CAPM fails to capture the momentum anomaly, leaving a significant portion of its excess return unexplained by market risk.

4.3 6 time-series tests of the Fama-French 5 factor mode

A time-series test of the Fama-French 5-Factor (FF5) model was conducted for each of the five industry portfolios. The following regression was estimated individually for

each industry $i \in \{1, 2, 3, 4, 5\}$, corresponding to the Oil, Banks, Textile, Toys, and Retail portfolios, respectively:

$$\begin{aligned} R_{i,t} - R_{f,t} = \alpha_i &+ \beta_{\text{Mkt-RF},i}(R_{M,t} - R_{f,t}) \\ &+ \beta_{\text{SMB},i}\text{SMB}_t \\ &+ \beta_{\text{HML},i}\text{HML}_t \\ &+ \beta_{\text{RMW},i}\text{RMW}_t \\ &+ \beta_{\text{CMA},i}\text{CMA}_t + \varepsilon_{i,t} \end{aligned} \tag{7}$$

Additionally, a 5-factor regression was performed on the Momentum factor (Mom_t) to test whether the Fama-French 5 factor model captures the momentum anomaly:

$$\begin{aligned} \text{Mom}_t = \alpha_{\text{Mom}} &+ \beta_{\text{Mkt-RF,Mom}}(R_{M,t} - R_{f,t}) \\ &+ \beta_{\text{SMB,Mom}}\text{SMB}_t \\ &+ \beta_{\text{HML,Mom}}\text{HML}_t \\ &+ \beta_{\text{RMW,Mom}}\text{RMW}_t \\ &+ \beta_{\text{CMA,Mom}}\text{CMA}_t + \varepsilon_{\text{Mom},t} \end{aligned} \tag{8}$$

Table 6: Fama-French 5-Factor Regression Results

Industry Portfolios	α_i	$\beta_{\textbf{Mkt-RF,i}}$	$\beta_{\textbf{SMB,i}}$	$\beta_{\textbf{HML,i}}$	$\beta_{\textbf{RMW,i}}$	$\beta_{\textbf{CMA,i}}$	**Adj** R^2
Oil	-0.1693	0.9739	-0.0273	0.3837	0.2179	0.3617	0.452610
	(-0.9971)	(23.7060)	(-0.4593)	(5.0025)	(2.7678)	(3.1590)	
Banks	-0.2421	1.1535	-0.0805	0.8460	0.0393	-0.3618	0.743414
	(-2.0908)	(41.1817)	(-1.9866)	(16.1777)	(0.7324)	(-4.6351)	
Txtls	-0.6157	1.1587	0.8386	0.5907	0.5740	0.0816	0.630170
	(-3.5202)	(27.3861)	(13.6961)	(7.4773)	(7.0784)	(0.6924)	
Toys	-0.5347	1.1193	0.7081	0.0069	0.5563	0.0684	0.594250
	(-3.0322)	(26.2383)	(11.4704)	(0.0867)	(6.8044)	(0.5755)	
Rtail	0.0585	0.9926	0.1412	-0.0665	0.3691	-0.0558	0.703130
	(0.5259)	(36.8841)	(3.6254)	(-1.3235)	(7.1570)	(-0.7436)	
Momentum Factor	$\alpha_{\textbf{Mom}}$	$\beta_{\textbf{Mkt-RF,Mom}}$	$\beta_{\textbf{SMB,Mom}}$	$\beta_{\textbf{HML,Mom}}$	$\beta_{\textbf{RMW,Mom}}$	$\beta_{\textbf{CMA,Mom}}$	
Momentum	0.7532	-0.1593	0.0066	-0.4726	0.1275	0.2986	0.088277
	(5.0029)	(-4.3742)	(0.1261)	(-6.9496)	(1.8261)	(2.9420)	

When examining Table 6, several notable findings emerge. Most prominently, the industry alphas for Banks, Textiles, and Toys become statistically significant, whereas under the CAPM they were statistically indistinguishable from zero. While this may initially seem counterintuitive as adding explanatory factors should theoretically better explain returns it is driven partly by a reduction in residual variance as can be seen when comparing the Adj R^2 values with the CAPM model. Specifically, the t-statistic is defined as $t(\hat{\alpha}) = \frac{\hat{\alpha}}{\text{SE}(\hat{\alpha})}$, where $\text{SE}(\hat{\alpha}) \propto \sigma_\varepsilon$. By adding SMB, HML, RMW, and CMA, the 5-factor model absorbs substantial unexplained variation, causing the residual standard deviation (σ_ε) to drop sharply. This shrinks the standard error in the denominator, which inflates the resulting t-statistic even when the point estimate $\hat{\alpha}$ remains unchanged. However it can be seen that the estimate $\hat{\alpha}$ also deviates further away from 0 in the industry portfolios.

This can be explained by omitted variable bias. When comparing the regressions of the CAPM model and the FF5 model the following can be seen:

$$\begin{aligned}\hat{\alpha}_{CAPM} &= \bar{y} - \hat{\beta}'_{CAPM}\bar{X}_{CAPM} \\ &= \hat{\alpha}_{FF5} + \hat{\beta}'_{FF5}\bar{X}_{FF5} - \hat{\beta}'_{CAPM}\bar{X_{CAPM}}\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\hat{\alpha}_{\text{CAPM}}] &= \alpha_{\text{FF5}} \\ &\quad + \beta_{\text{SMB}}\mathbb{E}[\text{SMB}] + \beta_{\text{HML}}\mathbb{E}[\text{HML}] \\ &\quad + \beta_{\text{RMW}}\mathbb{E}[\text{RMW}] + \beta_{\text{CMA}}\mathbb{E}[\text{CMA}] \\ &\quad + (\beta_{\text{Mkt, FF5}} - \beta_{\text{Mkt, CAPM}})\mathbb{E}[R_m - R_f]\end{aligned} \tag{9}$$

The larger magnitude of $\hat{\alpha}_{\text{FF5}}$ is explained by removing the omitted variable bias present in CAPM. In the CAPM opposing factor exposures cancel each other out, biasing $\hat{\alpha}_{\text{CAPM}}$ toward zero. Once this bias is controlled for in the FF5 model, the estimate unmasks itself and deviates farther away from zero.

Furthermore, the table indicates that the FF5 model fails to capture the momentum anomaly, as evidenced by a statistically significant intercept $\hat{\alpha}_{\text{mom}}$.

4.4 Momentum sorted portfolio returns

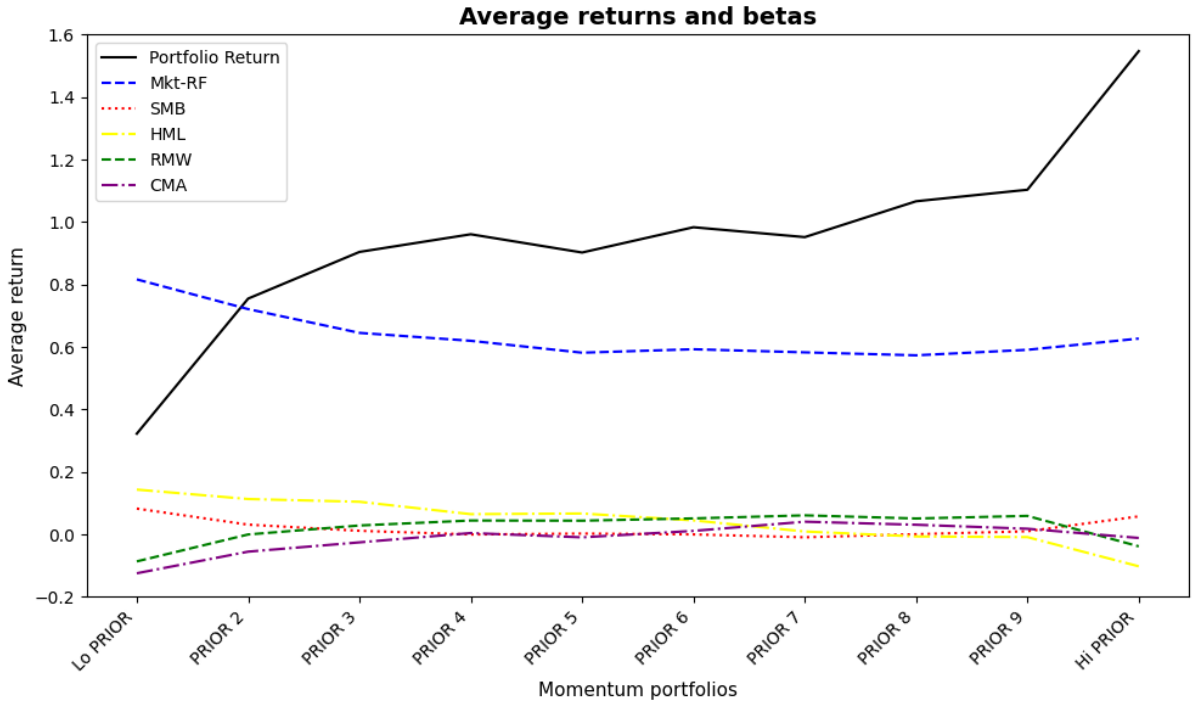


Figure 5: The graph illustrates the relative performance of ten momentum portfolios formed by sorting stocks into deciles based on their prior month returns. The LO PRIOR portfolio consists of the bottom 10% of assets with the lowest past returns, whereas the HI PRIOR portfolio represents the top 10% of past performers.

As shown in the graph, there appears to be a positive relationship between momentum and performance. The two lowest momentum portfolios yield the lowest expected excess returns, while the two highest momentum portfolios generate the highest. However, this upward trend in expected returns is not reflected by any of the factors. Instead, factor exposures remain flat across portfolios. This finding aligns with the empirical results reported in Table 6.

5 Stochastic discount factors

5.1 Model and Momentconditions

For this section we estimate the classic CAPM stochastic discount factor using monthly returns. We use the same five industry portfolios as in the rest of the report: Oil, Banks, Textiles, Toys and Retail. For this estimation we use the period from July 1963 until June 2026, which gives us 756 monthly observations.

Before estimating the model, we convert the returns to decimals and construct excess returns for the five industry portfolios. The normalized CAPM stochastic discount factor is

$$m_t = 1 - c(f_t - \mu),$$

where f_t is the market excess return, $\mu = E[f_t]$ and c is the slope parameter.
For each of the five industry portfolios we use the pricing condition,

$$E[m_t R^e_{i,t}] = 0.$$

We also use the additional condition

$$E[f_t - \mu] = 0.$$

This gives us six moment conditions in total and two parameters, μ and c. Since there are more moment conditions than parameters, the model is overidentified.

5.2 Two -step GMM estimator

The parameters are estimated by minimizing the GMM objective function,

$$Q_T(\theta) = \bar{g}_T(\theta)' W \bar{g}_T(\theta),$$

where $\bar{g}_T(\theta)$ is the average of the moment conditions and W is the weighting matrix.
We use two-step GMM. In the first step we use the identity matrix as weighting matrix,

$$W_1 = I_6.$$

This gives an estimate of 0.00597 for μ and 2.789 for c. After the first step we evaluate the moment conditions at these estimates and estimate their long-run covariance matrix using a Newey-West HAC estimator with 6 lags . This allows for heteroskedasticity and autocorrelation in the monthly time-series data.
The efficient weighting matrix for the second step is then,

$$W_2 = \hat{S}^{-1},$$

where $\hat{S}$ is the estimated Newey-West covariance matrix. We then minimize the GMM objective function again using this weighting matrix.
The results of both steps are shown in Table 7.

Table 7: GMM estimates of the CAPM SDF

	Step 1 estimate	Step 1 SE	Step 2 estimate	Step 2 SE
μ	0.00597	0.00166	0.00637	0.00164
c	2.789	1.076	3.571	1.028

The second-step estimates are the efficient GMM estimates. The estimate of μ is 0.00637, which corresponds to an average market excess return of around 0.637% per month. The estimate of the slope parameter c is 3.571. The reported standard errors are based on the GMM asymptotic variance matrix.

5.3 Model Tests

We then test the over-identifying restrictions using the Hansen J-test. The test statistic is

$$J = T\bar{g}_T(\hat{\theta}_2)' W_2 \bar{g}_T(\hat{\theta}_2).$$

Because we have six moment conditions and two estimated parameters, the test has $6 - 2 = 4$ degrees of freedom. We find a J-statistic of 4.777 and a p-value of 0.311. The null hypothesis is that the moment conditions are jointly valid. Since the p-value is larger than 0.05, we do not reject the null hypothesis. We therefore do not find evidence that the over-identifying restrictions are invalid.

Lastly, we test the restriction on the slope. Under the classic CAPM the restriction is

$$c = \frac{\mu}{\text{Var}(f_t)}.$$

Using our second-step estimate of μ and the variance of the market excess return, the restricted value of c is 3.202, while the estimated value from the second GMM step is 3.571.

We test this restriction using a Wald test. The Wald statistic is 0.556 and the p-value is 0.456. Since the p-value is larger than 0.05, we do not reject the null hypothesis that the slope restriction holds.

Overall, we fail to reject both the over-identifying restrictions and the slope restriction. For our five industry portfolios we therefore do not find evidence against the restrictions of the classic CAPM stochastic discount factor.

6 Discussion

Our three sets of results are not obviously consistent. We reject constant expected returns for all five industries, we do not reject the CAPM for those same five industries, and we do not reject the restrictions of the CAPM stochastic discount factor either, while neither the CAPM nor FF5 prices momentum. Below we argue that most of this tension disappears once the choice of test assets is taken into account.

6.1 Interpretation of the main results

The seasonality and the factor-model results concern different objects. Table 3 is about variation of $E[R_{i,t}]$ over time within an industry, while Section 4 is about the unconditional mean of $R_{i,t}$ relative to its factor exposure, and a model with constant β_i and a constant premium can match the mean while missing all variation around it. The reverse implication is the more awkward one for us: if expected returns move across months and across subperiods, then the constant-parameter specification of Section 4 is misspecified and $\hat{\alpha}_i$ and $\hat{\beta}_i$ are averages over a sample in which the true parameters moved. We would not overstate the seasonality itself either. The R^2 never exceeds 0.0214, and Table 4 shows that only Textiles rejects in all three subperiods. A pattern that relocates across industries over time is hard to separate from predictability that is traded away once it becomes known, which is what the efficient market hypothesis predicts in the first place.

The CAPM–FF5 comparison is the least intuitive result in the report. Four extra factors raise the adjusted R^2 for every industry (Banks 0.63 to 0.74, Textiles 0.46 to 0.63) but worsen the pricing errors: no CAPM alpha is significant, while three of five FF5 alphas are significant and negative, down to -0.62% per month for Textiles. Part of this is the mechanical fall in $\mathrm{SE}(\hat{\alpha})$, but the point estimates move as well. Textiles and Toys load heavily on SMB (0.84 and 0.71) and RMW (0.57 and 0.56), so FF5 predicts a premium that these industries did not earn; under the CAPM the same exposures sat in the residual and the omitted variable bias pulled $\hat{\alpha}_i$ toward zero. Fit and pricing are clearly not the same criterion, and the adjusted R^2 says nothing about whether a model prices an asset.

Momentum is the one result that is stable across specifications, with $\hat{\alpha}_{\mathrm{Mom}} = 0.71\%$ per month under the CAPM and 0.75% under FF5, both close to 9% annualised. Figure 5 shows why the extra factors do not help: average excess returns rise almost monotonically across the deciles

while the loadings stay flat, so there is a spread in means without a spread in exposures, and no model built from these five factors can price it. This is the same reading Cochrane (2011) gives of his Figure 6 for value. The significantly negative HML loading of the momentum factor (-0.47) only makes it worse, since value then pushes the predicted return in the wrong direction.

6.2 Test assets and power

The GMM results need the most care. Not rejecting ($J = 4.777$, $p = 0.311$, and $p = 0.456$ for the slope) is not evidence in favour of a model when the test has little power, and we believe power is the binding problem here. The five industries have market betas between 0.83 and 1.18 and mean monthly excess returns between 0.42% and 0.74%, so there is almost no cross-sectional spread to price, and any discount factor that prices the market will roughly price five near-market portfolios. The insignificant CAPM alphas in Table 5 and the insignificant J-statistic in Table 7 are then not two pieces of evidence but one, obtained twice from the same assets. Fama and French [1992] reject the CAPM on size and book-to-market sorts precisely because those sorts generate spread in average returns. Industry sorts do not.

The only asset in our report with real spread is momentum, and both models reject it. We did not include momentum-sorted portfolios in the moment conditions, and given the size of $\hat{\alpha}_{\text{Mom}}$ we would expect the over-identifying restrictions to be rejected if we had. The defensible statement is that the CAPM SDF is not rejected by these five industry portfolios, not that it is not rejected. The slope test is uninformative for the same reason: $\hat{c} = 3.571$ with a standard error of 1.028 gives a confidence interval of roughly 1.6 to 5.6, which is consistent with the restricted value of 3.202 and with a great deal else.

6.3 Limitations and further research

Apart from the test assets, the following qualifications apply.

- **Bandwidth choices.** Section 3 already shows that Oil no longer rejects at bandwidths of 6 and 12 months. We did not run the analogous check on the 6 lags used in the second GMM step.
- **Joint inference.** We test the five alphas one at a time while the industry residuals are positively correlated, so the individual t-tests overstate the joint evidence. A GRS test would be the appropriate procedure and would likely be less favourable to FF5.

6.3 Limitations and further research

- **Stationarity.** The asymptotics require stationarity and ergodicity, but the volatility of Toys falls visibly over the sample and the subperiod results are unstable. Time-varying betas, or simply estimating the factor models per subperiod, would be the natural check.
- **Portfolio construction.** The portfolios are value-weighted and SIC-based, so a few large firms drive the returns and the firms labelled Textiles in 1930 have little in common with those labelled Textiles in 2020. Part of what we read as an industry effect may be composition.
- **Data snooping and costs.** Momentum and calendar effects are among the most studied anomalies and we test them on the data that produced them; Holm corrects our own five tests, not the literature's. We also use nominal returns and ignore transaction costs, so we cannot claim the seasonal differences were tradeable.

The obvious extension is to re-estimate the SDF on assets that do have spread, for example the momentum deciles together with the industry portfolios, so that the over-identifying restrictions have something to reject, and to compare FF5 with a six-factor model including momentum on that same set of assets.

7 Conclusion

We analysed returns on five U.S. value-weighted industry portfolios, Oil, Banks, Textiles, Toys and Retail, from the Kenneth French data library, testing calendar predictability, the CAPM and the Fama-French five factor model in the time-series, and the classic CAPM stochastic discount factor by two-step GMM.

The five industries earn similar average returns of about 0.05% per day but differ in risk, with daily standard deviations of 2.14% for Toys against 1.15% for Retail, and all five are strongly leptokurtic and positively correlated. We reject the joint null of no month-of-the-year effects for every industry, and the rejections survive the Holm correction, so expected returns are not constant across calendar months. They are not stable either: across three equal subperiods only Textiles rejects in all three, Oil only in 1927–1959, and Toys and Retail only in 1993–2025, with R^2 between one and two percent throughout.

In the time-series tests the CAPM is not rejected for any industry, with insignificant alphas and strongly significant market betas, Toys the most cyclical ($\beta = 1.18$) and Oil the least ($\beta = 0.83$). FF5 raises the adjusted R^2 everywhere but produces significant negative alphas for Banks, Textiles and Toys, so the better fit comes with larger pricing errors. Neither model prices momentum: $\hat{\alpha}_{\text{Mom}} = 0.71\%$ per month under the CAPM and 0.75% under FF5, and the momentum deciles show a monotone spread in average returns against flat factor loadings. The GMM estimation gives $\hat{\mu} = 0.00637$ and $\hat{c} = 3.571$, with $J = 4.777$ ($p = 0.311$) and $p = 0.456$ for the restriction $c = \mu / \operatorname{Var}(f_t)$, so we find no evidence against the classic CAPM stochastic discount factor on these five portfolios.

We would read that last result narrowly. The industry portfolios all behave approximately like the market, with betas between 0.83 and 1.18 and little spread in mean returns, so our time-series and SDF tests have limited power against the CAPM and say as much about the test assets as about the model. The one set of assets in the report with genuine cross-sectional spread, the momentum-sorted portfolios, is rejected by both models. Our overall conclusion is therefore that the variation worth explaining, seasonal in the first half of the report and momentum in the second, is captured by neither the CAPM nor the Fama-French five factor model.

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A Descriptive Statistics

A.1 Industry Return Distributions

Table A.1: Descriptive Statistics for Daily Industry Portfolio Returns

Industry	N	Mean	Std. Dev.	Min	25th pct.	Median	75th pct.	Max	Skewness	Excess Kurt.	Shapiro–Wilk p
Oil	26,294	0.05	1.3552	-19.93	-0.56	0.05	0.65	19.12	-0.034	15.030	0.000
Banks	26,294	0.05	1.4779	-20.43	-0.51	0.05	0.62	19.06	0.213	19.369	0.000
Textiles	26,294	0.04	1.4227	-22.84	-0.54	0.05	0.64	19.43	-0.125	18.570	0.000
Toys	26,294	0.05	2.1385	-26.75	-0.87	0.00	0.92	39.74	0.529	24.882	0.000
Retail	26,294	0.05	1.1524	-17.97	-0.45	0.06	0.55	17.81	-0.029	12.969	0.000

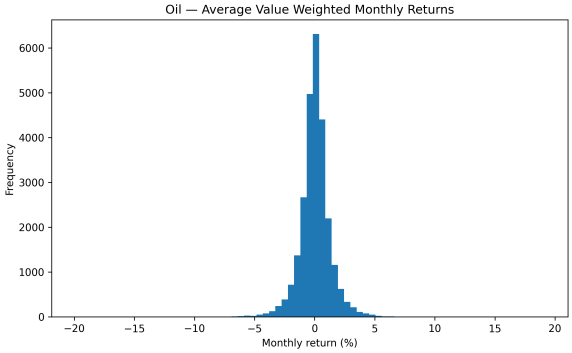
Notes: The descriptive statistics of the daily value-weighted industry returns. Excess kurtosis is measured relative to the normal-distribution value of zero. The Shapiro-Wilk test rejects normality for all five industries.

Table A.2: Descriptive Statistics for Monthly Industry Portfolio Returns

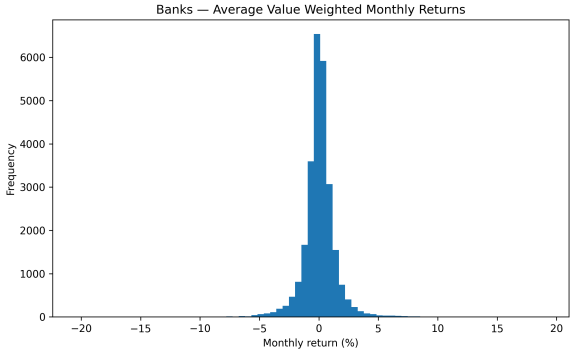
Industry	N	Mean	Std. Dev.	Min	25th pct.	Median	75th pct.	Max	Skewness	Excess Kurt.	Shapiro-Wilk p
Oil	756	1.03	6.0742	-34.65	-2.32	0.95	4.33	32.84	0.124	3.740	0.000
Banks	756	0.97	6.0558	-27.33	-2.30	1.04	4.77	24.98	-0.286	1.955	0.000
Textiles	756	0.93	7.6124	-36.02	-3.00	1.16	5.22	59.18	0.218	7.222	0.000
Toys	756	0.78	7.3271	-33.40	-3.82	1.02	5.51	27.70	-0.163	0.953	0.0017
Retail	756	1.10	5.4027	-28.89	-2.06	0.88	4.35	27.14	-0.088	1.939	0.001

Notes: These are the Monthly value-weighted industry returns. The excess kurtosis is measured relative to the normal-distribution value of zero. The Shapiro-Wilk test rejects normality for all five industries.

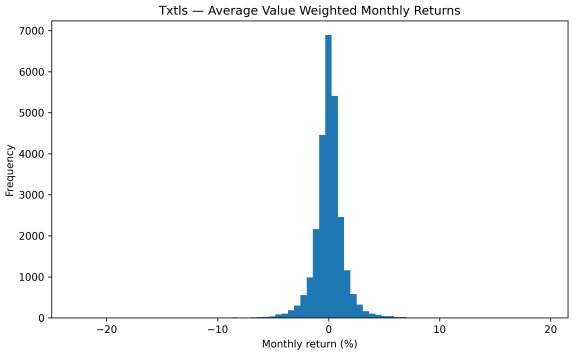
A.3 Distributions of Daily Industry Portfolio Returns



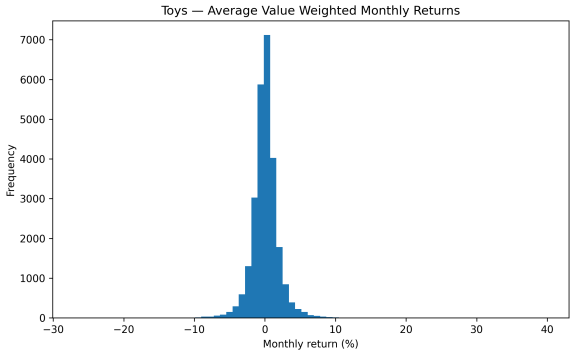
(a) Oil



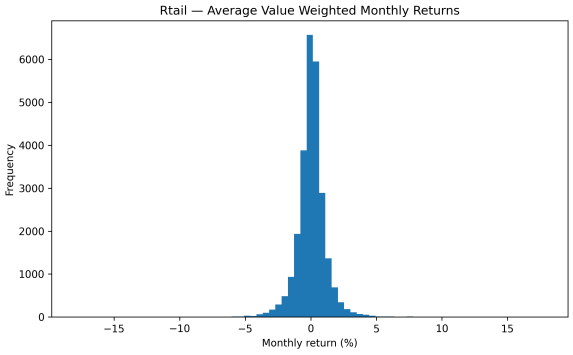
(b) Banks



(c) Textiles



(d) Toys



(e) Retail

A.2 Correlation Matrices of Daily and Monthly Industry Portfolio and Factor Returns

Table A.3: Correlation Matrix of Daily Industry Portfolio Returns

	Oil	Banks	Textiles	Toys	Retail
Oil	1.000	0.548	0.515	0.381	0.575
Banks		1.000	0.578	0.403	0.644
Textiles			1.000	0.440	0.610
Toys				1.000	0.490
Retail					1.000

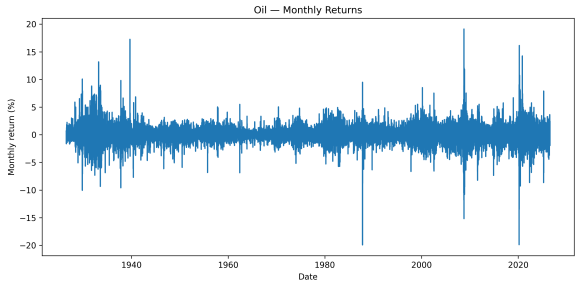
Notes: The correlations are calculated using daily value-weighted industry portfolio returns.

Table A.4: Correlation Matrix of Monthly Industry Portfolio and Factor Returns

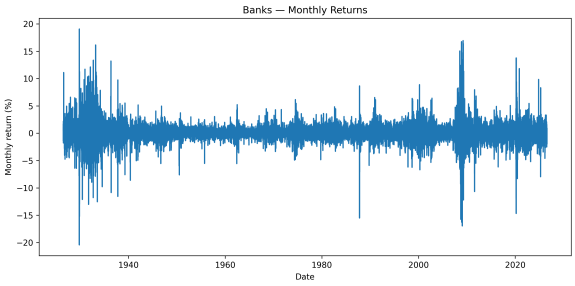
	Oil	Banks	Txtls	Toys	Rtail	Mkt-RF	SMB	HML	RMW	CMA	RF	Mom
Oil	1.000	0.498	0.411	0.405	0.358	0.609	0.146	0.131	-0.035	0.000	-0.016	-0.124
Banks		1.000	0.639	0.582	0.676	0.790	0.200	0.155	-0.101	-0.139	0.010	-0.294
Textiles			1.000	0.628	0.627	0.678	0.462	0.126	-0.052	-0.084	-0.024	-0.309
Toys				1.000	0.660	0.718	0.417	-0.104	-0.058	-0.240	-0.022	-0.203
Retail					1.000	0.823	0.249	-0.205	-0.035	-0.340	-0.011	-0.181
Mkt-RF						1.000	0.271	-0.210	-0.194	-0.356	-0.082	-0.165
SMB							1.000	0.016	-0.333	-0.076	-0.037	-0.081
HML								1.000	0.091	0.681	0.063	-0.193
RMW									1.000	0.012	-0.008	0.071
CMA										1.000	0.051	-0.020
RF											1.000	0.065
Mom												1.000

Notes: The correlations are calculated using monthly data for each portfolio.

A.4 Distributions of Daily Industry Portfolio Returns over Time



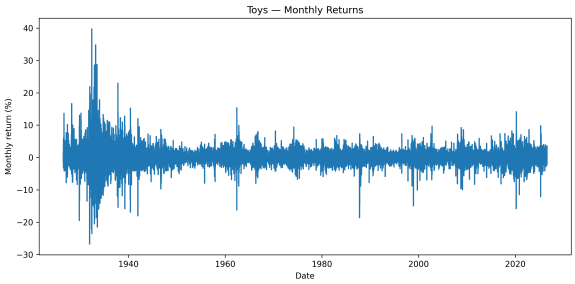
(a) Oil



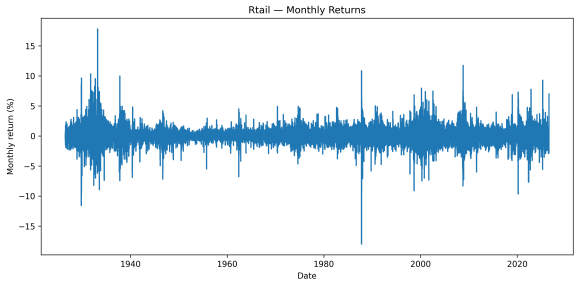
(b) Banks



(c) Textiles



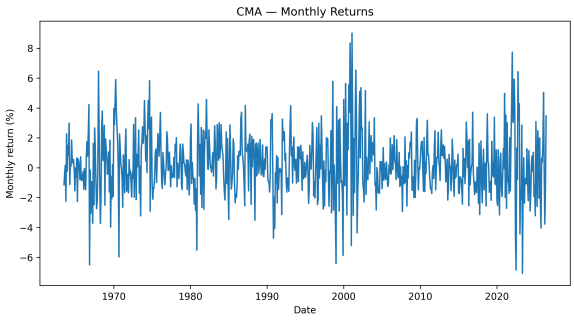
(d) Toys



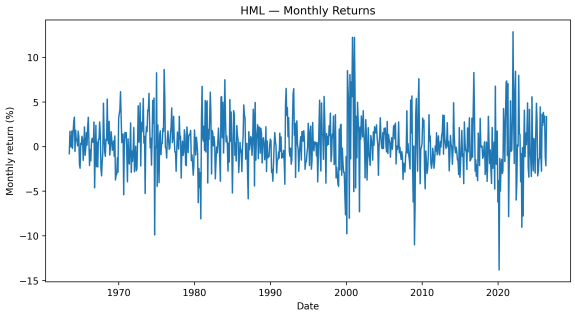
(e) Retail

Figure A.2: Daily Returns of the Five Industry Portfolios Over Time

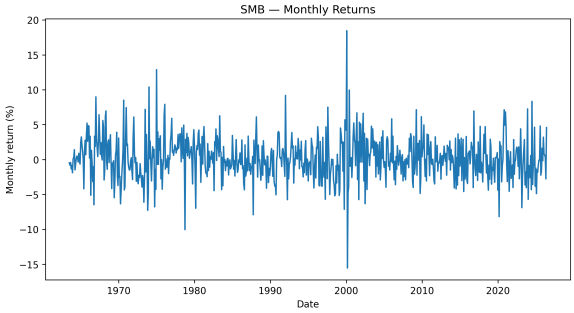
A.5 Monthly Returns of the Five Factor Portfolios Over Time



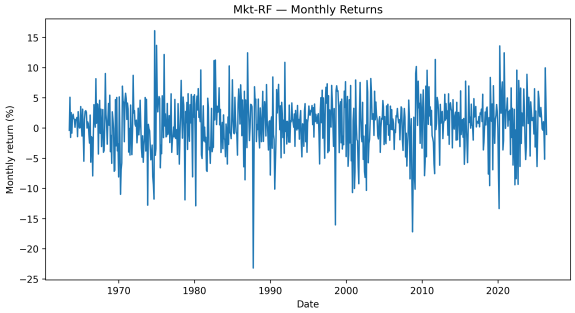
(a) Conservative Minus Aggressive



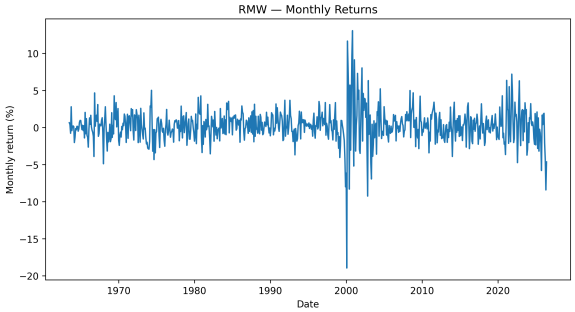
(b) High Minus Low



(c) Small Minus Big



(d) Market Excess Returns



(e) Robust Minus Weak

Figure A.3: Monthly Returns of the Five Factor Portfolios Over Time

B Predicatbility

This section gives us additional statistics on the month effect that we have checked.
On the following tables, you can find the estimation of each beta for each time period, together with the p-value and Wald statistic

Table B.5: Month-of-the-Year Regressions, 1927–1959

	Oil	Banks	Txtls	Toys	Rtail
January mean	0.397	4.628**	3.570***	1.715	1.239*
	(0.831)	(1.889)	(1.019)	(1.412)	(0.741)
February − January	-0.707	-2.356	-2.872**	0.475	-0.540
	(1.061)	(2.145)	(1.152)	(2.292)	(1.097)
March − January	1.271	-4.255*	-5.425***	0.977	-1.443
	(1.758)	(2.377)	(1.564)	(2.518)	(1.311)
April − January	2.494	-3.456*	-3.183	-0.912	0.185
	(1.797)	(1.929)	(1.957)	(2.177)	(1.318)
May − January	-0.305	-3.384	-2.837	-0.485	-0.588
	(1.181)	(2.974)	(1.855)	(2.050)	(1.404)
June − January	1.575	-2.292	-0.954	-1.476	1.212
	(1.396)	(1.724)	(1.684)	(2.122)	(1.302)
July − January	2.624*	-2.555	-0.182	0.148	1.763
	(1.471)	(2.471)	(2.146)	(2.993)	(1.492)
August − January	0.431	-1.228	-1.420	2.020	1.798
	(1.610)	(1.135)	(2.284)	(1.690)	(1.447)
September − January	-2.442	-5.888**	-4.117*	-2.011	-2.119
	(1.901)	(2.824)	(2.256)	(2.005)	(1.674)
October − January	0.903	-5.051**	-4.878***	-1.540	-1.818
	(1.191)	(2.573)	(1.646)	(2.970)	(1.452)
November − January	0.694	-3.117	-3.342**	1.420	0.281
	(1.334)	(2.587)	(1.633)	(2.480)	(1.422)
December − January	1.182	-4.153*	-3.055*	-4.908**	-1.589
	(1.295)	(2.288)	(1.638)	(2.130)	(1.286)
Joint Wald statistic	24.170	14.804	36.898	12.491	14.952
Joint p-value	0.0120	0.1916	0.0001	0.3279	0.1847
Holm p-value	0.0481	0.5541	0.0006	0.5541	0.5541
Observations	396	396	396	396	396
R^2	0.0344	0.0319	0.0358	0.0221	0.0345

Notes: The dependent variable is the monthly value-weighted industry return, expressed in percentage points. January is the omitted reference month, so the intercept is the average January return. The remaining coefficients measure the difference between the corresponding month's average return and the January average. Newey–West HAC standard errors with a 24-month bandwidth are reported in parentheses. The joint Wald test examines whether all eleven February–December coefficients are jointly equal to zero. Holm-adjusted p-values account for the five industry-level tests within the subperiod. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table B.6: Month-of-the-Year Regressions, 1960–1992

	Oil	Banks	Txtls	Toys	Rtail
January mean	1.116	2.319*	4.793***	4.086**	2.290*
	(1.115)	(1.218)	(1.197)	(1.829)	(1.273)
February − January	-0.868	-1.052	-2.258	-1.380	-0.246
	(1.208)	(1.556)	(1.462)	(2.057)	(1.244)
March − January	-0.323	-0.771	-3.403***	-1.993	0.236
	(1.202)	(1.435)	(1.258)	(1.760)	(1.389)
April − January	0.867	-1.231	-2.942**	-2.314	-0.643
	(1.060)	(1.519)	(1.442)	(1.871)	(1.700)
May − January	-0.536	-2.626*	-4.771***	-3.724*	-1.208
	(1.257)	(1.495)	(1.550)	(2.162)	(1.394)
June − January	-0.842	-1.670	-5.385***	-5.854***	-1.713
	(1.105)	(1.325)	(1.316)	(1.938)	(1.345)
July − January	0.166	-1.522	-3.454**	-4.240**	-2.032
	(1.525)	(1.662)	(1.416)	(2.148)	(1.838)
August − January	1.137	-0.371	-4.043**	-2.092	-0.789
	(1.370)	(1.973)	(1.805)	(2.265)	(1.945)
September − January	-1.132	-3.245*	-5.278***	-4.369*	-3.152*
	(1.312)	(1.758)	(1.535)	(2.554)	(1.756)
October − January	-0.122	-1.945	-5.392***	-5.619**	-2.115
	(1.836)	(1.762)	(1.857)	(2.458)	(1.830)
November − January	0.496	-0.785	-3.300*	-1.928	-0.162
	(1.805)	(1.732)	(1.903)	(2.384)	(2.018)
December − January	0.529	-0.328	-2.095	-3.533	-1.861
	(1.074)	(1.507)	(1.582)	(2.481)	(1.488)
Joint Wald statistic	17.280	19.319	37.991	20.441	19.814
Joint p-value	0.0999	0.0556	0.0001	0.0396	0.0480
Holm p-value	0.1586	0.1586	0.0004	0.1586	0.1586
Observations	396	396	396	396	396
R^2	0.0175	0.0249	0.0639	0.0439	0.0307

Notes: The dependent variable is the monthly value-weighted industry return, expressed in percentage points. January is the omitted reference month, so the intercept is the average January return. The remaining coefficients measure the difference between the corresponding month's average return and the January average. Newey–West HAC standard errors with a 24-month bandwidth are reported in parentheses. The joint Wald test examines whether all eleven February–December coefficients are jointly equal to zero. Holm-adjusted p-values account for the five industry-level tests within the subperiod. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

Table B.7: Month-of-the-Year Regressions, 1993–2025

	Oil	Banks	Txtls	Toys	Rtail
January mean	0.359	0.691	-0.496	0.699	0.628
	(1.105)	(1.283)	(1.248)	(1.117)	(0.898)
February − January	1.028	-0.667	1.554	0.920	0.078
	(1.324)	(1.313)	(1.522)	(1.473)	(1.296)
March − January	1.961	-0.245	0.439	-0.858	1.898
	(1.831)	(2.134)	(2.479)	(1.787)	(1.412)
April − January	2.863	1.617	4.608	1.043	0.579
	(1.928)	(1.962)	(3.198)	(1.982)	(1.321)
May − January	0.302	0.395	0.961	0.042	0.067
	(1.462)	(1.561)	(1.238)	(1.278)	(1.017)
June − January	-0.434	-1.321	-0.733	0.083	0.133
	(1.317)	(1.221)	(1.122)	(1.157)	(1.055)
July − January	-0.365	1.824	1.461	-0.265	1.004
	(1.057)	(1.546)	(2.508)	(1.773)	(1.202)
August − January	-0.748	-0.992	0.975	-0.852	-0.666
	(1.260)	(1.757)	(1.871)	(1.242)	(1.251)
September − January	0.230	-0.870	-2.205	-2.713*	-0.972
	(1.403)	(1.565)	(1.757)	(1.635)	(1.271)
October − January	0.721	1.320	-0.097	1.070	0.762
	(1.357)	(1.471)	(1.709)	(1.754)	(1.434)
November − January	1.286	1.723	3.144**	1.709	2.658**
	(1.374)	(1.578)	(1.351)	(1.381)	(1.058)
December − January	0.558	0.713	2.486	-2.646	-0.555
	(1.416)	(1.427)	(1.937)	(1.959)	(1.328)
Joint Wald statistic	16.416	20.944	30.917	25.119	38.844
Joint p-value	0.1264	0.0340	0.0011	0.0088	0.0001
Holm p-value	0.1264	0.0679	0.0045	0.0263	0.0003
Observations	396	396	396	396	396
R^2	0.0229	0.0312	0.0407	0.0381	0.0411

Notes: The dependent variable is the monthly value-weighted industry return, expressed in percentage points. January is the omitted reference month, so the intercept is the average January return. The remaining coefficients measure the difference between the corresponding month's average return and the January average. Newey–West HAC standard errors with a 24-month bandwidth are reported in parentheses. The joint Wald test examines whether all eleven February–December coefficients are jointly equal to zero. Holm-adjusted p-values account for the five industry-level tests within the subperiod. $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.10$.

C Python Code

C.1 Setting the stage

```
    import pandas as pd
    import numpy as np
    from scipy.stats import shapiro
    import matplotlib.pyplot as plt
    from pathlib import Path
    import io
    import statsmodels.api as sm
    import calendar

    data = Path(__file__).parent / "Data" / "49_Industry_Portfolios_Daily.csv"
    setting_outdir = Path(__file__).parent / "Setting the stage"
    predictive_outdir = Path(__file__).parent / "Predictability"

    histogram_dir = setting_outdir / "histograms"
    timeseries_dir = setting_outdir / "timeseries"

    setting_outdir.mkdir(exist_ok = True)
    predictive_outdir.mkdir(exist_ok = True)
    histogram_dir.mkdir(exist_ok = True)
    timeseries_dir.mkdir(exist_ok = True)



    lines = data.read_text(encoding="utf-8", errors = "replace").splitlines()
    section = next(i for i, line in enumerate(lines) if lines[i].strip() == "Average Value Weighted
Returns -- Daily" in line)
    header = section + 1
    end = next(i for i in range(header+1, len(lines)) if lines[i].strip() == "")


    text = "\n".join(lines[header:end])

    raw = pd.read_csv(io.StringIO(text))
    raw = raw.rename(columns={raw.columns[0]: "Date"})
    raw["Date"] = pd.to_datetime(raw["Date"].astype(str), format="%Y%m%d")

    industries = ["Oil", "Banks", "Txtls", "Toys", "Rtail"]

    for industry in industries:
        raw[industries] = raw[industries].apply(
            pd.to_numeric,
            errors="coerce"
        )

        raw[industries] = raw[industries].replace(
```

```
            [-99.99, -999],
            np.nan
        )

    start_date = pd.Timestamp("1926-07-01")

    daily_returns = (
        raw.loc[raw["Date"] >= start_date]
        .dropna()
        .set_index("Date")
        .sort_index()
    )

    industry_returns = daily_returns[industries]



    stats = pd.DataFrame({
        "N":                industry_returns.count(),
        "Mean (%)":         industry_returns.mean().round(2),
        "Std. dev. (%)":    industry_returns.std(ddof=1).round(4),
        "Min (%)":          industry_returns.min().round(2),
        "25th pct. (%)":    industry_returns.quantile(0.25).round(2),
        "Median (%)":       industry_returns.median().round(2),
        "75th pct. (%)":    industry_returns.quantile(0.75).round(2),
        "Max (%)":          industry_returns.max().round(2),
        "Skewness":         industry_returns.skew().round(3),
        "Excess kurtosis": industry_returns.kurt().round(3),
        "Shapiro (P-value)": industry_returns.apply(lambda x: shapiro(x.dropna()).pvalue).round(4),
    })

    corr = industry_returns.corr()

    print(stats)
    stats_path = setting_outdir / "Industry descriptive statistics.csv"
    corr_path = setting_outdir / "Industry correlation.csv"

    stats.to_csv(stats_path)
    corr.to_csv(corr_path)

    for industry in industries:
        plt.figure(figsize=(8, 5))
        plt.hist(
            industry_returns[industry].dropna(),
            bins=75,
            density=False
        )

        plt.xlabel("Monthly return (%)")
        plt.ylabel("Frequency")
```

```
        plt.title(f"{industry}  Average Value Weighted Monthly Returns")
        plt.tight_layout()
        plt.savefig(
            histogram_dir / f"{industry}_histogram.png",
            dpi=300,
            bbox_inches="tight"
        )

        plt.close()



        plt.figure(figsize=(10, 5))
        plt.plot(industry_returns[industry])
        plt.title(f"{industry}  Monthly Returns")
        plt.xlabel("Date")
        plt.ylabel("Monthly return (%)")



        plt.tight_layout()

        plt.savefig(
            timeseries_dir / f"{industry}_timeseries.png",
            dpi=300,
            bbox_inches="tight"
        )
        plt.close()


clean_path = setting_outdir / "selected_industry_returns_value_weighted_daily.csv"
daily_returns.to_csv(clean_path)

periods = {
    "Full": (None, None),
    "1926-1962": ("1926-07-01", "1962-12-31"),
    "1963-1977": ("1963-07-01", "1977-12-31"),
    "1978-2002": ("1978-01-01", "2002-12-31"),
    "2003-present": ("2003-01-01", None),
}

results = {
    "monday": [],
    "month": []
}

for period, (start, end) in periods.items():
    period_returns = industry_returns.loc[start:end]
    effects = [("monday", "Monday", period_returns.index.dayofweek == 0)]
```

```
            effects += [("month", calendar.month_abbr[m], period_returns.index.month == m) for m in range
    (1, 13)]

            for effect_type, effect_name, dummy in effects:
                X = pd.DataFrame({
                    "const": 1,
                    "Dummy": dummy.astype(int)
                }, index=period_returns.index)

                for industry in industries:
                    model = sm.OLS(period_returns[industry], X).fit(cov_type="HC0")

                    results[effect_type].append({
                        "Period":               period,
                        "Industry":             industry,
                        "Effect":               effect_name,
                        "Other mean (%)":       model.params["const"],
                        "Effect mean (%)":      model.params["const"] + model.params["Dummy"],
                        "Difference (%)":       model.params["Dummy"],
                        "t-stat":               model.tvalues["Dummy"],
                        "p-value":              model.pvalues["Dummy"]
                    })

        monday_results = pd.DataFrame(results["monday"]).round(4)
        month_results = pd.DataFrame(results["month"]).round(4)

        monday_results.to_csv(predictive_outdir / "monday_effect_subperiods.csv")
        month_results.to_csv(predictive_outdir / "month_effect_subperiods.csv")
```

C.2 Predictability

The following code is used to estimate monthly effects across 5 industries using appropriate standard errors, and later dividing the sample into parts to test persistence of time-variance of returns.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

from statsmodels.stats.diagnostic import (
    acorr_breusch_godfrey,
    het_arch
)

from statsmodels.stats.multitest import multipletests
```

C.2 Predictability

```
industries = [
    "Oil",
    "Banks",
    "Txtls",
    "Toys",
    "Rtail"
]


q3_data = q2_data.copy()

q3_data["Year"] = q3_data["Date"] // 100
q3_data["Month"] = q3_data["Date"] % 100

# Use the period reported in the assignment
q3_data = q3_data[
    q3_data["Year"].between(1927, 2025)
].copy()

# Keep only complete calendar years
months_per_year = (
    q3_data.groupby("Year")["Month"]
    .nunique()
)

complete_years = months_per_year[
    months_per_year == 12
].index

q3_data = q3_data[
    q3_data["Year"].isin(complete_years)
].copy()

print(
    "Question 3 sample:",
    q3_data["Date"].min(),
    "to",
    q3_data["Date"].max()
)

print("Observations:", len(q3_data))

diagnostic_results = []

for industry in industries:

    regression_data = q3_data[
```

```
        [industry, "Month"]
    ].dropna().copy()

    y = regression_data[industry]

    month_dummies = pd.get_dummies(
        regression_data["Month"],
        prefix="Month",
        dtype=float
    )



    # January is omitted
    dummy_columns = [
        f"Month_{month}"
        for month in range(2, 13)
    ]

    X = sm.add_constant(
        month_dummies[dummy_columns],
        has_constant="add"
    )

    # Preliminary OLS regression
    ols_model = sm.OLS(y, X).fit()

    # Breusch-Godfrey tests
    bg_12 = acorr_breusch_godfrey(
        ols_model,
        nlags=12
    )

    bg_24 = acorr_breusch_godfrey(
        ols_model,
        nlags=24
    )

    bg_36 = acorr_breusch_godfrey(
        ols_model,
        nlags=36
    )

    # ARCH-LM test
    arch_test = het_arch(
        ols_model.resid,
        nlags=12
    )

    diagnostic_results.append({
```

```
        "Industry": industry,
        "BG statistic (12)": bg_12[0],
        "BG p-value (12)": bg_12[1],
        "BG statistic (24)": bg_24[0],
        "BG p-value (24)": bg_24[1],
        "BG statistic (36)": bg_36[0],
        "BG p-value (36)": bg_36[1],
        "ARCH statistic": arch_test[0],
        "ARCH p-value": arch_test[1]
    })


diagnostic_results = pd.DataFrame(
    diagnostic_results
)

diagnostic_results["Serial correlation"] = np.where(
    diagnostic_results[
        [
            "BG p-value (12)",
            "BG p-value (24)",
            "BG p-value (36)"
        ]
    ].min(axis=1) < 0.05,
    "Yes",
    "No"
)

diagnostic_results["Heteroskedasticity"] = np.where(
    diagnostic_results["ARCH p-value"] < 0.05,
    "Yes",
    "No"
)



display(
    diagnostic_results.round(4)
)


joint_test_results = []


for industry in industries:

    regression_data = q3_data[
        [industry, "Month"]
    ].dropna().copy()
```

C.2 Predictability

```
# Dependent variable: raw industry return
y = regression_data[industry]

# Create calendar-month dummy variables
month_dummies = pd.get_dummies(
    regression_data["Month"],
    prefix="Month",
    dtype=float
)



# January is the omitted reference month
dummy_columns = [
    f"Month_{month}"
    for month in range(2, 13)
]

X = sm.add_constant(
    month_dummies[dummy_columns],
    has_constant="add"
)

# OLS coefficients with Newey-West HAC standard errors
final_model = sm.OLS(y, X).fit(
    cov_type="HAC",
    cov_kwds={"maxlags": 24}
)



# Joint test that all February-December effects equal zero
restriction_matrix = np.zeros(
    (
        len(dummy_columns),
        len(final_model.params)
    )
)

for row, variable in enumerate(dummy_columns):

    position = final_model.params.index.get_loc(
        variable
    )

    restriction_matrix[row, position] = 1

joint_test = final_model.wald_test(
```

```
        restriction_matrix,
        scalar=True
    )

    joint_test_results.append({
        "Industry": industry,
        "January mean": final_model.params["const"],
        "HAC SE January mean": final_model.bse["const"],
        "Wald statistic": np.asarray(
            joint_test.statistic
        ).item(),
        "Restrictions": len(dummy_columns),
        "p-value": np.asarray(
            joint_test.pvalue
        ).item(),
        "Observations": int(final_model.nobs),
        "R-squared": final_model.rsquared
    })

    # Save the individual monthly coefficients
    for month in range(2, 13):

        variable = f"Month_{month}"

        monthly_effect_results.append({
            "Industry": industry,
            "Month": month,
            "Effect relative to January":
                final_model.params[variable],
            "HAC SE": final_model.bse[variable],
            "t-statistic": final_model.tvalues[variable],
            "p-value": final_model.pvalues[variable]
        })


joint_test_results = pd.DataFrame(
    joint_test_results
)

monthly_effect_results = pd.DataFrame(
    monthly_effect_results
)

joint_test_results["Holm p-value"] = multipletests(
    joint_test_results["p-value"],
    method="holm"
)[1]

display(
```

```
    joint_test_results.round({
        "January mean": 3,
        "HAC SE January mean": 3,
        "Wald statistic": 3,
        "p-value": 4,
        "Holm p-value": 4,
        "R-squared": 4
    })
)

bandwidth_results = []

for bandwidth in [12, 24, 36]:

    bandwidth_period_results = []

    for industry in industries:

        regression_data = q3_data[
            [industry, "Month"]
        ].dropna().copy()

        y = regression_data[industry]

        month_dummies = pd.get_dummies(
            regression_data["Month"],
            prefix="Month",
            dtype=float
        )



        dummy_columns = [
            f"Month_{month}"
            for month in range(2, 13)
        ]

        X = sm.add_constant(
            month_dummies[dummy_columns],
            has_constant="add"
        )

        model = sm.OLS(y, X).fit(
            cov_type="HAC",
            cov_kwds={"maxlags": bandwidth}
        )

        restriction_matrix = np.zeros(
            (
```

```
                len(dummy_columns),
                len(model.params)
            )
        )

        for row, variable in enumerate(dummy_columns):

            position = model.params.index.get_loc(
                variable
            )

            restriction_matrix[row, position] = 1

        joint_test = model.wald_test(
            restriction_matrix,
            scalar=True
        )

        bandwidth_period_results.append({
            "Bandwidth": bandwidth,
            "Industry": industry,
            "p-value": np.asarray(
                joint_test.pvalue
            ).item()
        })

    bandwidth_pvalues = [
        result["p-value"]
        for result in bandwidth_period_results
    ]

    adjusted_pvalues = multipletests(
        bandwidth_pvalues,
        method="holm"
    )[1]

    for result, adjusted_pvalue in zip(
        bandwidth_period_results,
        adjusted_pvalues
    ):

        result["Holm p-value"] = adjusted_pvalue
        bandwidth_results.append(result)


bandwidth_results = pd.DataFrame(
    bandwidth_results
)
```

C.2 Predictability

```
bandwidth_table = bandwidth_results.pivot(
    index="Industry",
    columns="Bandwidth",
    values="Holm p-value"
)

display(
    bandwidth_table.round(4)
)

subperiods = {
    "1927--1959": (1927, 1959),
    "1960--1992": (1960, 1992),
    "1993--2025": (1993, 2025)
}

subperiod_results = []

for period_name, (start_year, end_year) in subperiods.items():

    period_data = q3_data[
        q3_data["Year"].between(
            start_year,
            end_year
        )
    ].copy()

    period_test_results = []

    for industry in industries:

        regression_data = period_data[
            [industry, "Month"]
        ].dropna().copy()

        y = regression_data[industry]

        month_dummies = pd.get_dummies(
            regression_data["Month"],
            prefix="Month",
            dtype=float
        )


        dummy_columns = [
            f"Month_{month}"
            for month in range(2, 13)
        ]
```

```
        X = sm.add_constant(
            month_dummies[dummy_columns],
            has_constant="add"
        )

        model = sm.OLS(y, X).fit(
            cov_type="HAC",
            cov_kwds={"maxlags": 24}
        )

        restriction_matrix = np.zeros(
            (
                len(dummy_columns),
                len(model.params)
            )
        )

        for row, variable in enumerate(dummy_columns):

            position = model.params.index.get_loc(
                variable
            )

            restriction_matrix[row, position] = 1

        joint_test = model.wald_test(
            restriction_matrix,
            scalar=True
        )

        period_test_results.append({
            "Period": period_name,
            "Industry": industry,
            "Wald statistic": np.asarray(
                joint_test.statistic
            ).item(),
            "p-value": np.asarray(
                joint_test.pvalue
            ).item(),
            "Observations": int(model.nobs)
        })

    period_pvalues = [
        result["p-value"]
        for result in period_test_results
    ]

    adjusted_pvalues = multipletests(
        period_pvalues,
```

```
        method="holm"
    )[1]

    for result, adjusted_pvalue in zip(
        period_test_results,
        adjusted_pvalues
    ):

        result["Holm p-value"] = adjusted_pvalue
        subperiod_results.append(result)


subperiod_results = pd.DataFrame(
    subperiod_results
)

subperiod_table = subperiod_results.pivot(
    index="Industry",
    columns="Period",
    values="Holm p-value"
)

display(
    subperiod_table.round(4)
)
```

C.3 Factor models

```
# Question 4a
CAPM = pd.merge(factors, returns, on='Date')

X = CAPM["Mkt-RF"]
X_with_const = sm.add_constant(X)

dependent_vars = {}
for col in ["Oil", "Banks", "Txtls", "Toys", "Rtail"]:
    dependent_vars[f"{col}"] = CAPM[col] - CAPM["RF"]

dependent_vars["Momentum"] = CAPM["Mom"]

results = []

for name, Y in dependent_vars.items():
    model = sm.OLS(Y, X_with_const).fit(cov_type='HAC', cov_kwds={"maxlags": 4})

    results.append(
        {
```

```
            "Portfolio/Factor": name,
            "Mean excess returns": dependent_vars[name].mean(),
            "Alpha": model.params["const"],
            "Alpha t-stat": model.tvalues["const"],
            "Beta": model.params["Mkt-RF"],
            "Beta t-stat": model.tvalues["Mkt-RF"],
            "R-squared": model.rsquared_adj,
        }
    )

capm_results = pd.DataFrame(results)

print(capm_results.to_string(index=False))
# Question 4b

X2 = CAPM[["Mkt-RF","SMB","HML","RMW","CMA"]]
X2_with_const = sm.add_constant(X2)

results2 = []

for name, Y in dependent_vars.items():
    model2 = sm.OLS(Y, X2_with_const).fit(cov_type='HAC',cov_kwds={"maxlags": 4})

    results2.append(
        {
            "Industry portfolio": name,
            "Intercept": model2.params["const"],
            "Intercept t-stat": model2.tvalues["const"],
            "Beta Mkt-RF": model2.params["Mkt-RF"],
            "Beta Mkt-RF t-stat": model2.tvalues["Mkt-RF"],
            "Beta SMB": model2.params["SMB"],
            "Beta SMB t-stat": model2.tvalues["SMB"],
            "Beta HML": model2.params["HML"],
            "Beta HML t-stat": model2.tvalues["HML"],
            "Beta RMW": model2.params["RMW"],
            "Beta RMW t-stat": model2.tvalues["RMW"],
            "Beta CMA": model2.params["CMA"],
            "Beta CMA t-stat": model2.tvalues["CMA"],
            "R-squared": model2.rsquared_adj,
        }
    )

capm_results2 = pd.DataFrame(results2)

print(capm_results2.to_string(index=False))
# Question 4c
momentum_portfolios=pd.read_csv('10_Portfolios_Prior_12_2.csv', skiprows=10, engine='python', on_bad_lines
    ='skip')
momentum_portfolios = momentum_portfolios.rename(columns={momentum_portfolios.columns[0]: 'Date'})
```

```
momentum_portfolios.columns = momentum_portfolios.columns.str.strip()
momentum_portfolios = momentum_portfolios.iloc[:1195].reset_index(drop=True)
momentum_portfolios = momentum_portfolios.apply(pd.to_numeric, errors='coerce')
momentum_portfolios['Date'] = pd.to_datetime(
    momentum_portfolios['Date'].astype(str).str.split('.').str[0],
    format='%Y%m',
    errors='coerce'
)
X3= CAPM[["Mkt-RF","SMB","HML","RMW","CMA"]]
X3_const = sm.add_constant(X3)
momentum_portfolios = momentum_portfolios[momentum_portfolios['Date'].isin(CAPM['Date'])].reset_index(drop
    =True)

factor_mean=X3.mean()
dependent_vars2 = {}
for col in momentum_portfolios.columns[1:]:
    dependent_vars2[f"{col}"] = momentum_portfolios[col] - CAPM["RF"]

results3=[]
for name, Y in dependent_vars2.items():
    model3 = sm.OLS(Y, X3_const).fit()

    results3.append(
        {
            "Momentum portfolio": name,
            "Beta Mkt-RF": model3.params["Mkt-RF"],
            "Beta SMB": model3.params["SMB"],
            "Beta HML": model3.params["HML"],
            "Beta RMW": model3.params["RMW"],
            "Beta CMA": model3.params["CMA"],
        }
    )
factor_cols = ["Mkt-RF", "SMB", "HML", "RMW", "CMA"]

factor_effects = {f: [] for f in factor_cols}
for row in results3:
    factor_effects["Mkt-RF"].append(row["Beta Mkt-RF"] * factor_mean["Mkt-RF"])
    factor_effects["SMB"].append(row["Beta SMB"] * factor_mean["SMB"])
    factor_effects["HML"].append(row["Beta HML"] * factor_mean["HML"])
    factor_effects["RMW"].append(row["Beta RMW"] * factor_mean["RMW"])
    factor_effects["CMA"].append(row["Beta CMA"] * factor_mean["CMA"])

portfolio_names = list(momentum_portfolios.columns[1:])
avg_returns = [momentum_portfolios[col].mean() for col in portfolio_names]
x_axis = np.arange(len(portfolio_names))
styles = {
    'Mkt-RF': ('blue', '--'),
    'SMB': ('red', ':'),
    'HML': ('yellow', '-.'),
```

```
    'RMW': ('green', '--'),
    'CMA': ('purple', '-.')
}
plt.figure(figsize=(10, 6))

# Plot Portfolio Average Excess Returns
plt.plot(x_axis, avg_returns, color='black', label='Portfolio Return', linewidth=1.5)

# Plot each Factor Effect Line
for f in factor_cols:
    color, lstyle = styles[f]
    plt.plot(x_axis, factor_effects[f], label=f, color=color, linestyle=lstyle)

plt.title('Average returns and betas', fontsize=14, fontweight='bold')
plt.xlabel('Momentum portfolios', fontsize=11)
plt.ylabel('Average return', fontsize=11)
plt.xticks(ticks=x_axis, labels=portfolio_names, rotation=45, ha='right')
plt.ylim(-0.2, 1.6)
plt.legend(loc='upper left')

plt.tight_layout()
plt.show()
```

Listing 1: Python Example

C.4 Stochastic Discount Factor

The following code is used for the CAPM stochastic discount factor analysis, including the two-step GMM estimation, standard errors, Hansen J-test and slope restriction test.

```
import pandas as pd
import numpy as np
from scipy.optimize import minimize
from scipy.stats import chi2

# Load data
df = pd.read_csv('49_Industry_Portfolios.csv', skiprows=11,
                 engine='python', on_bad_lines='skip')

df = df.rename(columns={df.columns[0]: 'Date'})
df.columns = df.columns.str.strip()

keep_cols = ['Date', 'Oil', 'Banks', 'Txtls', 'Toys', 'Rtail']
df = df[keep_cols]
df = df.iloc[:1200].reset_index(drop=True)
```

C.4 Stochastic Discount Factor

```
factors = pd.read_csv('F-F_Research_Data_5_Factors_2x3.csv', skiprows=4,
                      engine='python', on_bad_lines='skip')

factors = factors.rename(columns={factors.columns[0]: 'Date'})
factors.columns = factors.columns.str.strip()
factors['Date'] = pd.to_numeric(factors['Date'], errors='coerce')
factors = factors[factors['Date'].between(100000, 999999)]
factors = factors[['Date', 'Mkt-RF', 'RF']].reset_index(drop=True)

df['Date'] = pd.to_numeric(df['Date'], errors='coerce').astype('Int64')
factors['Date'] = pd.to_numeric(factors['Date'], errors='coerce').astype('Int64')

data = pd.merge(df, factors, on='Date', how='inner')

# Construct excess returns
return_cols = ['Oil', 'Banks', 'Txtls', 'Toys', 'Rtail', 'Mkt-RF', 'RF']

data[return_cols] = data[return_cols].apply(pd.to_numeric, errors='coerce')
data = data.dropna(subset=return_cols).reset_index(drop=True)
data[return_cols] = data[return_cols] / 100

industries = ['Oil', 'Banks', 'Txtls', 'Toys', 'Rtail']

for industry_name in industries:
    data[industry_name + '_excess'] = data[industry_name] - data['RF']

Re = data[['Oil_excess', 'Banks_excess', 'Txtls_excess',
           'Toys_excess', 'Rtail_excess']].to_numpy()

f = data['Mkt-RF'].to_numpy()

T = len(f)
N = Re.shape[1]

mu_start = np.mean(f)
c_start = 1.0
theta_start = np.array([mu_start, c_start])

# GMM moment conditions
def moments(theta):
    mu, c = theta
    m = 1 - c * (f - mu)
    moment_market = f - mu
    moment_assets = m[:, None] * Re
    return np.column_stack((moment_market, moment_assets))

# Step 1 GMM
W1 = np.eye(6)
```

C.4 Stochastic Discount Factor

```
def gmm_objective(theta, W):
    g_bar = moments(theta).mean(axis=0)
    return g_bar @ W @ g_bar

result1 = minimize(gmm_objective, theta_start, args=(W1,), method='BFGS')

theta1 = result1.x
mu1, c1 = theta1

# Newey-West covariance matrix
g1 = moments(theta1)

def newey_west_S(g, q):
    T = g.shape[0]
    S = (g.T @ g) / T

    for j in range(1, q + 1):
        weight = 1 - j / (q + 1)
        Gamma_j = (g[j:].T @ g[:-j]) / T
        S += weight * (Gamma_j + Gamma_j.T)

    return S

q = int(np.floor(4 * (T / 100) ** (2 / 9)))
S1 = newey_west_S(g1, q)
W2 = np.linalg.inv(S1)

# Step 2 GMM
result2 = minimize(gmm_objective, theta1, args=(W2,), method='BFGS')

theta2 = result2.x
mu2, c2 = theta2

# Standard errors
def gbar(theta):
    return moments(theta).mean(axis=0)

def numerical_jacobian(func, theta, h=1e-6):
    theta = np.asarray(theta, dtype=float)
    D = np.zeros((6, 2))

    for j in range(2):
        step = np.zeros(2)
        step[j] = h
        D[:, j] = (func(theta + step) - func(theta - step)) / (2 * h)

    return D

D1 = numerical_jacobian(gbar, theta1)
```

```
A1 = D1.T @ W1 @ D1
B1 = D1.T @ W1 @ S1 @ W1 @ D1
V1 = np.linalg.inv(A1) @ B1 @ np.linalg.inv(A1) / T
se1 = np.sqrt(np.diag(V1))

g2 = moments(theta2)
S2 = newey_west_S(g2, q)

D2 = numerical_jacobian(gbar, theta2)

A2 = D2.T @ W2 @ D2
B2 = D2.T @ W2 @ S2 @ W2 @ D2
V2 = np.linalg.inv(A2) @ B2 @ np.linalg.inv(A2) / T
se2 = np.sqrt(np.diag(V2))

print("Step 1:")
print("mu =", mu1, "SE =", se1[0])
print("c =", c1, "SE =", se1[1])

print("Step 2:")
print("mu =", mu2, "SE =", se2[0])
print("c =", c2, "SE =", se2[1])

# Hansen J-test
g_bar2 = moments(theta2).mean(axis=0)

J_stat = T * (g_bar2 @ W2 @ g_bar2)
J_df = 6 - 2
J_pvalue = chi2.sf(J_stat, J_df)

print("Hansen J-statistic:", J_stat)
print("Degrees of freedom:", J_df)
print("P-value:", J_pvalue)

# Slope restriction
var_f = np.var(f, ddof=0)

restriction = c2 - mu2 / var_f
gradient = np.array([-1 / var_f, 1])

var_restriction = gradient @ V2 @ gradient
se_restriction = np.sqrt(var_restriction)

z_stat = restriction / se_restriction
p_value_slope = chi2.sf(z_stat ** 2, df=1)

print("Estimated c:", c2)
print("CAPM restricted c:", mu2 / var_f)
```

```
print("Difference:", restriction)
print("SE difference:", se_restriction)
print("z-statistic:", z_stat)
print("p-value:", p_value_slope)
```

D AI Statement

During the preparation of this work, the student used ChatGPT in order to assist with spelling, punctuation, grammar, syntax, clarity, and referencing. The following parts of the assignment were affected by AI tool usage: Management summary, Setting the Stage, Predictability, Factor models, Stochastic discount factos, Discussion, Conclusion and the Appendix. After using this tool/service, Kelvin Schouten evaluated the validity of the tool's outputs, including the references where applicable, and edited the content as needed. As a consequence, Kelvin Schouten, Antonio Harley Sardenberg,Rik Crijns, Victor Kense and Walid Afasi takes full responsibility for the content of this work.